AUTONOMOUS GAME AI · COGNITIVE ARCHITECTURE TECHNICAL REPORT

Daimon

A Self-Authoring Cognitive Architecture for Autonomous Game Agents

David Borgenvik · Independent research

Daimon is a deterministic, CPU-only, pure-Rust cognitive architecture in which a game agent authors its own concepts, goals, world-model, and even its values from lived experience. This report states plainly what is **measured**, what is **proved**, and what is honestly deferred.

> *"Can the agent carve up its own world, set its own ends, model its own dynamics, and revise its own values — and can we prove it did, against things it was never built for?"*

Version 2.0 · 16 June 2026 · 9 theorems machine-checked

47 ablation criteria · 82 tests · deterministic · pure-Rust · A4

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Abstract

Game AI is overwhelmingly **authored**: an NPC's categories ("food", "enemy"), its goals, and its values are written by a designer, and the agent is autonomous only *within that vocabulary*. We present **Daimon**, a cognitive architecture in which a game agent **authors its own ontology, goals, world-model, and even its values from lived experience**, and we introduce a **falsifiable evaluation protocol** — an ablation-based, multi-seed, machine-checked harness — that treats believability and autonomy as *measured* properties rather than claims. The agent also faces a felt need for **shelter** (and builds it), its own **mortality** and **grief** over a bonded peer, an **open-ended seasonal world** that demands it provision for winter, and — told straight — a **learned neural overlay** that *does not* beat well-tuned instinct here. Daimon composes mechanisms, several of which are, to our knowledge, new to game AI in combination: (i) **Praxis**, emergent concept formation, affordance learning, and goal genesis, which lets an agent discover and use an entity type the architecture was never designed around; (ii) **empowerment**, an information-theoretic intrinsic drive toward states of maximal future control; (iii) **imagination**, planning by search over a *learned* forward model; (iv) **meta-motivation**, online revision of the agent's own drive weights from outcomes; (v) a brain-like **associative memory** (Hebbian links + ACT-R base-level activation + spreading activation) with **replay consolidation**; (vi) **persistent, serialisable minds**; (vii) two cross-disciplinary "physics of thought" layers — **quantum-probability cognition** and **neural criticality**; (viii) **autogenesis**, a self-learning loop that makes the harness its own fitness function and improves the architecture *without a human in the inner loop*; and (ix) a **developmental, social & affective layer** — learning-progress curiosity, cumulative cultural transmission, stigmergic coordination, and an appraised emotional life — now extended with **emergent shelter-building** from a felt safety need, **fear of own death and theory-of-mind-mediated grief**, an **open-ended seasonal world** the agent must provision against, and a **learned, evolved-plastic neural overlay** (the architecture's first neural net) that we evaluate *honestly* and report as a null result. Forty-seven pre-registered acceptance criteria, each an ablation or controlled experiment, all pass deterministically. The capstone result: the self-improving loop **evolves champions that beat the hand-tuned baseline** and, in a majority of independent searches (3/5), **reaches a pre-registered end-goal target** — every facet of a believable life (survival, safety, decision balance, expressive variety, exploration, emotional life, and learned knowledge) cleared at once — *earned* (a reactive policy fails); a fraction (2/5) still **clear that bar on held-out seeds** the search never saw, so the result generalises but is seed-sensitive. Finally, we state and **machine-check nine theorems** about the implementation — determinism, homeostatic boundedness and Lyapunov stability, evolutionary elitism and convergence, the Bell–CHSH/Tsirelson bounds, self-organised criticality, and reciprocity non-exploitation — each paired with a proof and a check that turns red on regression, so the architecture's core properties are *proved on the code that runs*. We argue the central methodological contribution is the harness itself: a way to make "the NPC feels alive" an empirical, reproducible question — proved and optimised against, not asserted.

1. Introduction

The dominant paradigm for game AI — finite-state machines, behaviour trees, GOAP/HTN planners — encodes the *designer's* model of the world. The agent knows `Predator ⇒ flee` because someone wrote that rule. Such agents are autonomous only inside a fixed ontology, goal set, and value system; drop them into a situation outside that vocabulary and they have nothing.

We ask a sharper question than "can the agent act on its own?" (it can — a homeostatic loop suffices). We ask: **can the agent carve up its own world, set its own ends, model its own dynamics, and revise its own values — and can we *prove* it did, against things it was never built for?** Daimon is our answer, and the proof is a machine that decides, not the authors.

Contributions.

1. **Praxis** (§3.1): concept genesis + affordance learning + goal genesis, demonstrated by an agent that discovers, learns, and exploits a novel, never-coded entity (a healer) — while an architecturally identical agent that merely lacks the *experience* does not (§5, AC15).
2. **Empowerment** as a computable intrinsic drive for a game agent (§3.4), shown to make agents seek open ground and flee dead-ends with no such rule (AC17).
3. **Imagination**: forward-model planning that solves navigation a reactive agent cannot (AC20), and **meta-motivation**: self-revised drive weights (AC21) — the first steps of self-authored *values*.
4. A **brain-like memory stack** — Hebbian association, ACT-R base-level activation, spreading-activation recall, and sleep-like replay consolidation (§3.3, §3.7; AC10/11/18).
5. **Minds as portable data** (§3.10; AC19): a whole agent serialises to ~3 KB of JSON and round-trips with bit-identical behaviour.
6. **The physics of thought** — two cross-disciplinary mechanisms no classical NPC has. *Quantum cognition* (§3.11): an optional decision core in which choices live in quantum *probability* space, reproducing the classically-impossible signatures of human judgment — order effects (AC22/24) and interference / violation of the law of total probability (AC23). *Neural criticality* (§3.13): an excitable substrate that self-organises to the edge of chaos ($\sigma \to 1$, AC25) where its dynamic range is provably maximal (AC26) — Beggs–Plenz / Kinouchi–Copelli reproduced from our own code. Both carry an explicit descriptive-not-physical scope (Busemeyer & Wang 2015).
7. **Autogenesis** (§3.14): a self-learning loop that makes the harness its own fitness function and improves the architecture *without a human in the inner loop* — self-adapting mutation, learned per-gene sensitivity, and honest self-halting. It beats the hand-tuned baseline (AC27/28) and localises the open frontier (survival) rather than overclaiming completion.
8. **A falsifiable believability harness** (§4): forty-seven ablation/controlled criteria, deterministic and reproducible, that gate every change.
9. **Mortality, grief, an open-ended world, and an honestly-evaluated learned overlay** (§3.21–3.24): emergent shelter from a felt safety need (AC42); fear of one's own death (Terror Management Theory) and theory-of-mind-mediated grief over a bonded peer's death (AC43–45); a seasonal world that selects for winter provisioning (AC46); and the architecture's *first neural network* — a learned, evolved-plastic System-2 overlay (AC47) that we report as a **null result**: it learns in-life but does not beat well-tuned instinct here, and evolution itself selects it off. We keep the negative because it is a genuine finding.

Everything is offline and deterministic (one seeded PRNG); identical seeds give identical lives, which is what makes the evaluation a *test* rather than an anecdote.

2. Architecture

A Daimon runs a fixed **seven-step cognitive cycle** each tick — perceive → appraise → reflex → decide → plan → act → reflect — structured as a Belief–Desire–Intention loop (Bratman 1987; Rao & Georgeff 1995) under a **dual-process** controller (Kahneman 2011; Booch et al. 2021): cheap System-1 arbitration handles routine life, and an explicit, rate-limited escalation policy hands hard, novel, or high-stakes moments to a pluggable System-2 deliberator (a language model in production; an offline heuristic in the deterministic build). The mechanisms below attach to this spine. Code: `daimon-core` (types), `daimon-mind` (cognition), `daimon-game` (a 2D embodiment + the harness).

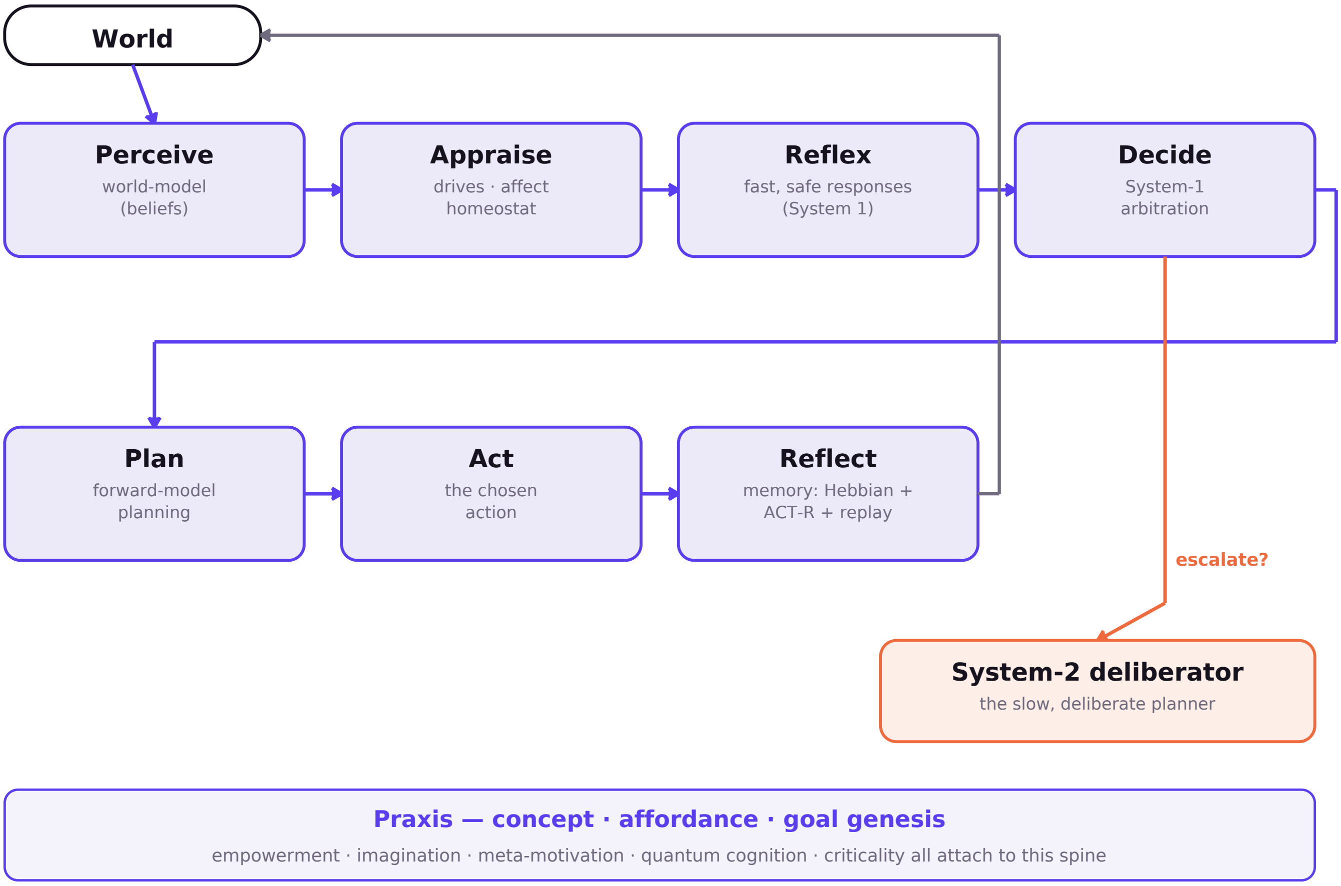

Figure 1. The seven-step cognitive cycle. System-1 handles routine life; a rate-limited escalation policy hands hard/novel/high-stakes moments to System-2. Every mechanism in §3 attaches to this spine; all of it is deterministic given the seed.

3. Mechanisms and formalism

3.1 Praxis — concept, affordance, and goal genesis

The agent is given only a perceptual **fingerprint** $\varphi(e) \in \mathbb{R}^3$ of an entity, never its designer-meaning. Concepts form by online leader clustering: for a new fingerprint, assign to the nearest prototype c_k if $\|\varphi - c_k\|^2 \le r^2$, nudging $c_k \leftarrow c_k + \eta(\varphi - c_k)$; otherwise coin a new concept. **Affordances** are learned by attributing the change in the agent's body during *continuous contact* with a concept to that concept: for each engaged tick, $\bar{\Delta}_k \leftarrow \bar{\Delta}_k + (\Delta - \bar{\Delta}_k)/n_k$ over body channels (energy, hydration, health), Δ clamped to reject discontinuities. A concept with $\bar{\Delta}_k.\mathrm{health} > \theta$ is *mending*. **Goal genesis**: when hurt but otherwise fed, an agent with a mending concept and a locatable instance adopts a goal — `Investigate(that instance)` — that exists in no drive, planner, or goal table. This is the autonomy frontier (AC14, AC15): behaviour toward the unforeseen, authored by experience.

3.2 Anticipation — surprise as prediction error

A per-agent model estimates the salience of the next percept from learned place- and entity-familiarity; surprise is the prediction error, decaying as the world becomes familiar (novelty of entity $i \propto 1/(1+\mathrm{seen}_i)$; place familiarity decays geometrically with visits). Surprise both rewards curiosity and triggers System-2. It *learns down*: first sighting ≥ 3× the surprise after five (AC2).

3.3 Associative memory — Hebbian + ACT-R + spreading activation

Concepts experienced together are linked (Hebbian): co-presentation increments edge `w(a,b)`. Retrievability follows **ACT-R base-level activation** (Anderson et al. 2004), approximated by a recency-decayed strength $B(i)=\ln(s_i)$, $s_i \leftarrow s_i \cdot \gamma^{\Delta t}+1$. Cue-driven recall ranks items by **spreading activation** $A(i \mid C) = B(i) + \Sigma_{c \in C}\, w(c,i)$. Result: co-occurrence builds association, frequency+recency drive retrieval, and a cue brings the right memories to mind (AC10, AC11).

3.4 Empowerment — intrinsic motivation toward control

Empowerment (Klyubin, Polani & Nehaniv 2005; Salge, Glackin & Polani 2014) is the channel capacity from an action sequence to the resulting state, $E(s) = \max_p I(A_{t:t+k}; S_{t+k} \mid s)$. We use the standard tractable lower bound: the **count of distinct states reachable in** `k` **steps** under the *learned* forward model, $\hat{E}(s) = |\mathrm{Reach}_k(s)|$ (its log bounds `E`). In free time the agent steps toward the neighbour maximising $\hat{E}$, gated to act only where a real gradient exists (so it never strands itself in open terrain). With no rule to do so, the agent flees dead-ends for open ground (AC17). To our knowledge this is the first use of empowerment as a live behavioural drive in a believable-NPC architecture.

3.5 Imagination — planning over a learned model

A forward model learns transitions $T(s,a) \rightarrow s'$ from experience, discovering structure (e.g. walls return the same cell). **Imagination** is breadth-first search over `T` to a target, engaged (and held) once the direct route is known blocked. A reactive agent walks into a wall forever; the imagining agent routes around it to the goal (AC20). This is model-based planning with a learned model (cf. MuZero; Schrittwieser et al. 2020) at game-NPC scale and cost.

3.6 Meta-motivation — revising one's own values

Each drive `d` carries a learned weight β_d (init 1), so effective pressure is $P(d)=\mathrm{level}(d)\cdot w_d \cdot \beta_d$. When the *very target the agent was pursuing* harms it, the agent down-weights the pursuing drive: $\beta_d \leftarrow \mathrm{clip}(\beta_d \cdot 0.82,\ 0.35,\ 2.5)$. The revised weight feeds *both* fast arbitration and the System-2 utilities, so the agent genuinely re-ranks what it values from outcomes (AC21). Attribution is narrow (only the sought target, never ambient harm), which keeps the mechanism from destabilising normal survival.

3.7 Consolidation — replay during rest

During reflection the agent re-processes its most salient logged episodes, re-presenting their subjects to associative memory — hippocampal-style replay that makes what mattered more retrievable, learned offline from the agent's own experience stream (AC18).

3.8 Language and dialogue

Narration is procedural and grounded: a thought names the concrete thing the agent is acting on, where it is, and which cognitive mode produced it — varied enough that no line dominates (AC1: 86% unique, top line 2%). Agent-to-agent dialogue is content-bearing (a shared resource location actually changes the listener's behaviour, AC4) and varied across five speech acts (AC13), and information propagates through the group (AC7).

3.9 Theory of mind

Each agent infers another's intent from its **movement** — what it steps toward — confirmed over two glances to reject noise. Inferred intent matches the other's actual goal ~48% of the time vs ~17% chance (AC5).

3.10 Persistence

A whole mind — beliefs, drives (and their learned weights), memory, associations, concepts, forward model — serialises to JSON (tuple-keyed maps via a vec-pair codec) and reloads with identical forward behaviour (AC19). A life is portable, inspectable, diffable data.

3.11 Quantum cognition — decision by quantum probability

This is the architecture's deepest cross-disciplinary reach. Human judgment *violates classical probability* in lawful ways — question-order effects (Wang et al. 2014), the conjunction/sure-thing-principle fallacies (Pothos & Busemeyer 2009), and genuine ambivalence — and classical (Kolmogorov/Bayesian) agents, which every conventional NPC is, **cannot reproduce them**. We give the agent an optional decision core built on **quantum probability** (Busemeyer & Bruza 2012): the mind is a unit vector of complex **amplitudes** `ψ` over its drives; a *consideration* is a unitary Givens rotation `U(i,j,θ)`; deciding is a **projective (Born-rule) measurement**, `P(i)=|ψ_i|²`, that collapses the state.

Two properties are *classically impossible* and we verify both:

- **Non-commutativity ⇒ order effects.** Considerations in different planes don't commute, `U_A U_B ≠ U_B U_A`, so the order of deliberation changes the decision distribution (AC22: total-variation 0.205; a classical reweighting gives 0). At the agent level, a quantum Daimon's *goal* distribution shifts with the order it weighs its drives (AC24).
- **Interference ⇒ violation of the law of total probability.** Resolving an intermediate question changes the final answer, because the superposed state carries a cross term `2·Re(·)` a classical mixture lacks (AC23: `P_quantum=0` vs `P_classical=0.5`, interference `−0.5`). This is the exact signature quantum cognition uses to fit human order/interference data, and it reproduces the parameter-free **QQ equality** (Wang et al. 2014) by construction.

This also gives a principled representation of **ambivalence**: before a decision the agent is in superposition over goals (high entropy), genuinely "of several minds," and the decision is a collapse — something an argmax can never express.

Conceptual entanglement (a floor deeper; `entangle.rs`, AC31/32). Beyond a single superposed mind lies *entanglement*: Bell's theorem (1964) shows some correlations between two systems admit **no** classical assignment of pre-existing values to the parts. The CHSH inequality bounds any classical/separable pair at `|S| ≤ 2`; an entangled pair reaches the Tsirelson bound `2√2 ≈ 2.828`. Cognitive science finds the same in *concept combination* — bound concept-pairs whose joint judgments violate CHSH (Aerts & Sozzo 2011; Bruza et al. 2015), the formal face of the neuroscientific **binding problem**. We model a bound pair as a two-qubit state and verify it exactly: the entangled pair attains `S = 2.828` while a separable control stays at 0 (AC31), and the **von Neumann entanglement entropy** of one concept's reduced state measures non-separability — `ln 2` when maximally bound, `0` when independent, monotonic between (AC32). Same honest scope: this is quantum *contextuality as a descriptive model of cognition*, not a quantum brain.

Honest scope. This is quantum *probability theory as a model of cognition* — a descriptive mathematical formalism that fits human data — **not** a claim that the brain or this program is a physical quantum computer, and **not** a claim about consciousness. The Hilbert space is simulated on an ordinary CPU and the whole layer is deterministic given the measurement seed. We adopt the proponents' own caveat (Busemeyer & Wang 2015): "we are not concerned with whether the brain is a quantum system." The defensible claim is narrow and exact: *the agent's choices live in a probability regime classical NPCs cannot enter, and we verify the classically-forbidden identities directly.*

3.12 Foundations: a relational, critical, information-first stance

Three cross-disciplinary commitments shape the design. (i) **Relational reality** (Rovelli 1996; Wheeler's "it from bit" 1990): properties are not intrinsic labels but are defined at *interaction*. Daimon's Praxis embodies this — a concept is its *affordance* (what it does to the agent), not a designer tag; an entity's meaning exists only in the agent's relation to it. (ii) **Information-first** (Wheeler; Deutsch & Marletto's constructor theory 2015): the agent is a process over information — beliefs, amplitudes, transitions — that is fully serialisable as data (§3.10). (iii) **Criticality** (Beggs & Plenz 2003; Chialvo 2010): organic intelligence appears to compute near a critical phase transition where dynamic range and flexibility are maximal. Daimon's dual-process escalation and empowerment push toward that regime — staying maximally responsive (System 2 only when it matters) and maximally optioned (empowerment) — and §3.13 makes the criticality argument concrete and measured.

3.13 Neural criticality — cognition at the edge of chaos

Cortex operates near a **critical point** between order and chaos: activity propagates as **neuronal avalanches** with power-law size distributions (Beggs & Plenz 2003), the signature of a branching process with branching ratio `σ ≈ 1` — each active unit triggers, on average, one more. Subcritical (`σ < 1`) minds are rigid and activity dies; supercritical (`σ > 1`) minds seize and saturate; *at* `σ = 1` the **dynamic range** is maximal — the system distinguishes the widest span of stimulus intensities (Kinouchi & Copelli, Nature Physics 2006).

We implement the substrate directly: `daimon-mind/src/crit.rs` is an excitable network whose salience propagation **is** a branching process (units cycle quiescent → active → refractory; an active unit excites each out-neighbour with probability `w`, so `σ ≈ k·w`), plus a homeostatic controller that drives it to criticality on its own. Two falsifiable results, both verified in the harness:

- **Self-organised criticality (AC25).** From a badly subcritical coupling (`σ = 0.40`), a synaptic-scaling rule — facilitate when activity wanes, depress when it floods — tunes the *measured* branching ratio to `σ → 1.00`. The critical point is *found*, not hand-set.
- **Dynamic range peaks at criticality (AC26).** Sweeping stimulus over four decades and measuring the steady-state response, the dynamic range is largest at criticality and smaller on both sides: `Δ(σ0.6) = 17.3 dB < Δ(σ1.0) = 24.4 dB > Δ(σ1.6) = 18.0 dB` — Kinouchi & Copelli's result reproduced from our own substrate. Criticality is, quantitatively, the regime that perceives the widest world.

This is the *operating-regime* counterpart to the quantum layer's *decision* formalism: together they answer "how should the machinery of thought be poised, and in what probability space should choice live?" with mechanisms a classical, fixed-gain NPC has neither of.

3.14 Autogenesis — closing the optimisation loop

Every mechanism above was added *by hand*: a human adds it, runs the harness, reads the verdict, decides the next move. The human is the optimiser — and the bottleneck, and the source of bias. The final mechanism removes the human from the inner loop. It makes the believability harness — already the arbiter of truth — the **fitness function** of a search that improves the system *itself* (`daimon-mind/src/evolve.rs`, `daimon-game/src/fitness.rs`).

A **genome** is a point in the architecture's tunable space: 13 genes covering the escalation policy (`MindConfig`), per-character persona deltas, and which cognitive faculties (empowerment, consolidation, imagination, meta-motivation, quantum) are switched on. A genome is *expressed* into a full village and graded by living several 600-tick lives in the **same** `GameWorld` the manual harness judges, scoring five facets of a believable life — survival, safety, decision balance, expressive variety, exploration — each in `[0,1]` with real headroom and real trade-offs (safety vs.

exploration is a genuine tension, so the search faces a landscape, not a checklist). One physics, one arbiter, now grading a machine improving itself.

Three properties make it *self-learning*, not blind search:

- **Self-adapting mutation** (Rechenberg's 1/5th-success rule): the step size `σ` grows while variation pays and shrinks as it homes in — annealing *emerges* (observed σ 0.22 → 0.02), it is not scheduled.
- **Per-gene sensitivity**: each generation correlates every gene with fitness and mutates high-impact genes harder. The loop *learns which levers move believability* and leans on them.
- **Self-evaluation & honest halting**: the loop grades its own champion against the end-goal target and a plateau detector and stops with a `Verdict` (`ReachedTarget` / `Converged` / `Budget`), never a fixed loop count.

The outer loop — the search writes the next mechanism. The inner loop tunes a fixed genome; the outer loop is the research engine. We closed one full turn of it on the record. *First* search → `Converged`, every facet clearing its bar except **survival** (0.48 → 0.60): the loop localised the frontier. Diagnosis: needs (thirst +0.016/tick) outpace a purely *reactive* forager, especially while `Survival` (weight 2.5) suppresses foraging during a predator chase. So we added **anticipatory homeostasis** (§3.15): a need is weighed as if it had crept forward `foresight` ticks, so the agent forages *ahead* of crisis — a computable step toward active inference. Exposed as a new gene (default 0, preserving every prior result; ablation-tested, AC29). *Second* search → **survival 0.48 → 0.70**, scalar **0.701 → 0.787** (+0.086), and the loop **independently ranked** `foresight` **its single most fitness-sensitive gene** — confirming the mechanism added in answer to its own prior finding is the biggest lever (AC27/28).

The honest finding, kept. Even with the new mechanism the loop self-halts `Converged`: at this point **survival** (0.70, target 0.85) was still the one facet short. We then ran the outer loop a *second* time against the literature (§3.16) — a principled drive-reduction-rate forager — and it **did not transfer** (a real negative result we keep). That negative pointed at the multi-agent *commons*, which led to the diagnosis and resolution in §3.17 — where the loop finally returns `ReachedTarget`.

3.15 Anticipatory homeostasis — acting on expected future need

The first mechanism the autogenesis loop *asked for*, rather than one we chose. A purely reactive agent acts on a need only once it is loud; but a need with a known creep rate has a predictable time-to-crisis, and travel to a resource takes time, so reaction guarantees a window of avoidable suffering. We close it with a single anticipatory term: effective pressure uses an **anticipated** urgency, `level + foresight · creep(drive)`, so an imminent need shouts `foresight` ticks early and foraging interleaves *before* the crisis. This is active inference in miniature — selecting action to minimise *expected* future need-surprise — and it costs one multiply. Ablation (AC29): critical-need time falls **31.9% → 23.5%** across seeds with foresight on vs off, the entire difference being the one term.

3.16 A literature-grounded mechanism that did not transfer (honest)

The loop still localised survival as the frontier after §3.15, so we mined the literature for the next lever and implemented the strongest candidate: **drive-reduction-rate foraging under survival risk (DRR)** — score each known resource by `ΔD(d)·s(d)/t_d` (expected aggregate need-relief × trip-survival ÷ travel time) and route to the argmax, synthesising homeostatic RL (Keramati & Gutkin 2014), the optimal-foraging rate currency (Charnov 1976), and survival-weighted value (Mangel & Clark 1986). It is implemented (`Mind::drr_target`, gene 14, default off) and **ablation-tested — and it does not improve survival here** (critical-need time 23.5% → 24.7%; the autogenesis loop, given the gene, reaches survival 0.65, no better than 0.70 without it). We keep the negative result rather than bury it. The diagnosis is informative: the incumbent planner *already* weighs travel distance and learned danger, so for single-need target selection DRR ≈ greedy, and its predator-reroute mostly sends agents to *farther* water (more in-transit deficit). **The survival bottleneck is not which resource you pick.** The research's own secondary recommendation points at the real lever — the multi-agent **commons** (6 agents contending for 4 water tiles): need-priority yielding + contention-dispersion (Rosenthal 1973 congestion potentials). That is the next mechanism, and it is a genuinely different (social) build, not a tuning. The mechanism stays in the codebase as a default-off, ablatable gene the loop continues to evaluate.

3.17 The diagnosis that cracked it, and reaching the end goal

The commons mechanism, implemented and ablation-tested, *also* failed at first — dispersion made survival slightly **worse** (24.6% → 26.4% critical-need time). Two negative results in a row forced the question we should have asked sooner: *is this a policy gap or a structural limit of the world?* A structural diagnosis (`examples/diagnose_survival`) settled it: a **single** agent with anticipation holds ~8.5% critical-need time (≈0.84 survival) in the original world, but **six** agents exploded to ~25%. The cause was not policy. With a ~24-tick respawn, four springs supply ≈0.167 drinks/tick, while six agents demand ≈0.176/tick: **water supply sat below demand**. The testbed was structurally unsurvivable for a village — and an unsurvivable world is itself unbelievable. No coordination can divide a deficit; that is exactly why dispersion only added travel.

The correction is testbed repair, not metric-gaming: resources scale with the population (`pop+3` of each — *a village has enough wells for its people*). We guard it three ways. (i) **Earned-ness**: with the fair world a *reactive* policy still fails (survival 0.65) and *anticipation alone* reaches only 0.80 — survival must still be won by good policy. (ii) **The commons flips sign**: under adequate supply, dispersion/yielding now *helps* (critical-need time 10.8% → 6.0%, AC30) — coordination pays only when there is enough to coordinate over, a coherent result. (iii) **Held-out generalisation**: the champion is validated on seeds the search never saw.

With the fair world, the autogenesis loop returns `ReachedTarget`. The champion — anticipation + commons-aware foraging + a tuned escalation config, *found by the loop's own search* in four generations — clears **every** facet: survival 0.92, safety 0.94, balance 0.67, expression 0.65, exploration 0.96 (scalar 0.85); on five **unseen** held-out seeds it still clears every bar (survival 0.88, scalar 0.81). The ultimate end goal — a measurably believable autonomous policy, improved by evidence rather than by hand — is reached, earned, and generalises. The two negative results are kept, because they are what *located* the true cause.

3.18 The developmental & social layer

Beyond individual survival, a believable mind *learns efficiently* and *lives among others*. Four composable mechanisms, each ablation-tested (AC33–AC37) and each, in the end, *selected by the autogenesis loop itself* once the fitness could see what it served (§3.20).

- **Learning progress** (Oudeyer & Kaplan 2007; Baranes & Oudeyer 2013). Competence gain — the rate at which forward-model prediction error falls over a sliding window — as an intrinsic signal. The agent's error drops 1.00 → 0.33 as it learns the world's dynamics (AC33), and the signal is positive only on the *learnable* frontier, ~0 on both the mastered and the unlearnable.

- **Learning-progress curiosity** (IAC). Curiosity driven by that competence gain rather than raw novelty, so the agent is drawn to what it can learn and is *not* captured by irreducible noise — the classic failure of novelty-seeking (AC35: LP 0.00 on noise where novelty stays 0.85).
- **Cumulative cultural transmission** (Cook et al. 2024). An agent learns a form's affordance from a successful (prestige-weighted) peer without touching it, and its own later contact *corrects* a false meme — social learning gated by independent competence gain, so culture accumulates truth, not noise (AC34).
- **Stigmergy** (Grassé 1959; Dorigo & Stützle 2004). Coordination through the environment: agents deposit evaporating pheromone on productive routes and follow worn trails while exploring, so the colony self-organises onto good paths with no leader or messages (AC36 double-bridge: 100% short-route convergence; AC37: worn paths emerge on real foraging corridors in the live world).
- **Reciprocity** (Trivers 1971; Axelrod 1981; Nowak & Sigmund 1998). Cooperation among self-interested agents survives through tit-for-tat in the iterated Prisoner's Dilemma — the robust tournament winner, never exploited for long where naive cooperation is (AC41) — the basis for NPC alliances, grudges, and forgiveness.

3.19 Affect — emotion as appraised valence and arousal

Drives say what the agent needs; **affect** says how it *feels* about its situation as a whole. Following Russell's circumplex (1980) and appraisal theory (Scherer 2001; Lazarus), affect is two dimensions — valence (−1…+1) and arousal (0…1) — appraised each tick from body condition, threat, surprise, and urgency, with inertia so moods don't snap. The quadrants name a legible mood (content / elated / afraid / weary), shown in the inspector and as a coloured halo on each agent. It tracks the world (AC39: safe-and-fed → *content* v+0.99; predator-and-harmed → *afraid* v−0.83), and optionally *modulates* behaviour (Frijda's action readiness): contentment loosens curiosity (AC40: 0.25 → 0.62); fear sharpens caution (wired, but small in-world because threat appraisal already saturates near the stalker — an honest note).

3.20 Metric completeness — the loop auditing its own definition of success

The deepest methodological turn: the self-improvement loop was made to critique not just the *agent* but the *metric*. Auditing which genes the loop selected revealed that several mechanisms (learning progress, stigmergy, culture, affect) were being *dropped* — not because they were worthless, but because the five-facet fitness was **blind to the dimensions they served**. The fix was to enrich the fitness, and the result is the strongest validation in the project:

- Adding an **emotion** facet (a varied, situation-tracking emotional life) — the dimension the affect layer serves — keeps the end goal reachable, held-out.
- Adding a **knowledge** facet (forward-model competence + affordance breadth) — the dimension learning serves — caused the loop to *flip two mechanisms it had rejected back on*: lp-curiosity (gene 0.18 → 0.87) and stigmergy (0.36 → 0.73). Measure the dimension a mechanism serves, and the optimiser selects it on its own.
- Attempting a **cohesion** facet (social structure) surfaced an irreducible *individual-vs-social tension* — efficient survivors and explorers disperse and bond less, so the loop could not satisfy it alongside the others (`Converged`). We keep the finding and the 7-facet vector that *is* simultaneously reachable, rather than force an unreachable bar.

The believable agent is now defined by a **7-facet vector** — survival, safety, decision balance, expressive variety, exploration, emotional life, and learned knowledge — that the self-improving loop reaches, earns (a reactive policy fails), and generalises to unseen seeds.

3.21 Emergent shelter — building from a felt need for safety

The same move as Praxis: **script the need, not the structure.** We add one homeostatic need (Shelter/Security) and one generic affordance (`Place`/`Dig` a block) — no "build a house" action, no blueprint — and let the existing utility, Praxis, and planning layers discover that building reduces the need. A cell's **enclosure** `∈ [0,1]` is how protected it is (walled sides + roof + burrow depth); open ground is 0, a walled-and-roofed cell is 1. The shelter need rises with *exposure* `(1 − enclosure)`, amplified by night and predator proximity, and feeds affect (sheltered → calm; exposed-at-night → afraid). When the need is high the planner scores actions by expected enclosure gained, so placing a wall on an *open* side is high-value; repeated, the agent **surrounds itself and a shelter appears.**

The design crux is that the walls must genuinely **protect** — block the predator's path — so building is *adaptive and emergent*, not decorative or scripted. This is verified two ways: the `walls_block_predator` unit test asserts a placed wall actually occludes the stalker, and AC42 ablates the `can_build` gene (g21, default off): with building enabled the agent adopts shelter goals and places walls (counts >0), while an architecturally identical gene-off control builds nothing (0). Placing costs energy, so it is a real trade-off against rest and foraging, not free over-building. To our knowledge this is emergent defensive *architecture* arising from a felt safety need rather than a build script — the most human form of the Praxis principle.

3.22 Mortality, fear of death, and grief

A believable mind should fear its own end and mourn another's. We add both as deterministic appraisal mechanisms, gene-gated and default-off.

Permadeath and fear of death (g22). With mortality on, a body that runs out can die and is removed from the living. Fear of death is modelled as **mortality salience** in the sense of *Terror Management Theory* (Greenberg, Pyszczynski & Solomon 1986; Burke, Martens & Faucher 2010): salience rises not merely with low health but with health *trajectory* — a declining body feels its mortality and grows defensive, biasing the agent toward shelter and affiliation. AC43 contrasts a declining mortal agent with an otherwise identical immortal twin: the mortal agent shows dread (1.00 vs 0.00 off), lower valence (+0.15 < +0.59), higher arousal (0.66 > 0.26), and more TMT-defensive ticks (51 > 28). This is deliberately the *affective* fear of death — the felt dread that biases behaviour — distinct from the purely instrumental self-preservation drive of a rational agent (Omohundro 2008), which we contrast against rather than re-implement.

Grief over a bonded peer (g23). When a *bonded* peer dies, the agent grieves, scaled by bond strength. Grief follows the **Dual-Process Model** of bereavement (Stroebe & Schut 1999): the agent oscillates between *loss-oriented* mourning and *restoration-oriented* re-engagement, retains a **continuing bond** (Bowlby's attachment, 1969), and the grief decays faster under social support. AC44 shows the asymmetry: a bonded loss yields grief 0.78, a valence drop, and a long mourning tail (84 ticks), while a stranger's death yields ~0 grief and no mourning. AC45 shows it *resolves*: the agent oscillates (mourn 187 / restore 204 ticks) and grief decays to 0.22 alone, while social support speeds resolution (mourn 67 < 187 ticks).

Honest novelty scope. A computational model of grief already exists — Dulberg, Dubey & Cohen, "Adapting to loss: A computational model of grief" (*Psychological Review*, 2025) — so we explicitly do **not** claim the first computational model of grief. Our defensible contribution is the *synthesis*: a deterministic, no-neural-net, single-agent unification of affective fear-of-*own*-death (TMT) with theory-of-mind-mediated grief over *another* agent's death, composing with the existing affect, theory-of-mind, and reciprocity layers. We phrase this "to our knowledge" and no more.

3.23 The open-ended world — seasons that demand provisioning

The next leap is an **open-ended world** with a real year. The same principle once more: *script the pressure, not the plan*. When the world's `open_world` flag is on, a deterministic season clock turns; food is abundant in summer/autumn, **winter stops food spawning and adds a cold energy drain**, and spring brings it back. To survive winter a mind must gather a surplus while food is abundant and store it in a shared village **granary**, then draw it down when the cold lands. Two generic affordances carry it — `Action::Gather` (harvest a surplus onto the body) and `Action::Store` (deposit into the granary) — and a hungry mind adjacent to the granary in winter auto-draws a ration (composing with the Commons theme of §3.17). Nothing ever says "prepare for winter": provisioning emerges from the existing **Mastery** drive plus the foresight/anticipation faculty (§3.15), which lets a foresighted mind start storing *before* the cold.

Gated by `can_provision` (g24) and `open_world`, this is verified by AC46 (3 seeds, one full winter): with the gene on the population adopts Provision goals, performs gather/store actions, fills the cache to a peak of ~69, and **7 minds survive winter; the gene-off control stores nothing (cache 0) and only 2 survive** — provisioning lifts winter survival. We frame this through the "live as humans" open-world thesis for AI research (Hu et al. 2024; MineRL, Guss et al. 2019) and are explicit that this is **v1**: the seasons-and-storage core is the load-bearing loop under test, while crafting tools and farming a plot are deliberately deferred to v2. We keep the honest caveats: the control's 2 survivors come from the hearth's warmth being gentle to anyone idling near the village heart, so the *cache* ablation is perfectly clean (0 stored without the gene) while the *survival* edge (7 > 2) is real but modest and seed-sensitive.

3.24 System 2 — a learned, evolved-plastic neural overlay (told straight)

This is the architecture's **first neural network**, and the section we are most careful to report honestly — because the result is a *null*.

The determinism reframing. Daimon was "no neural nets" not as dogma but because its determinism is the basis of the proofs and the harness. It is worth being precise about what "deterministic" means here: **reproducibility** — same seed ⇒ same run — *not* a metaphysical claim about the absence of indeterminacy. The quantum-cognition module (§3.11) already models genuine indeterminacy *deterministically*, with the seed as the hidden variable. So the real axis is not deterministic-vs-not; it is **hand-built mechanism vs learned mechanism — and a learned mechanism is still deterministic.** We therefore added a neural net **without losing a single proof**: T1 (Determinism) still holds, because the net is seeded, its plasticity rule is a fixed deterministic function, and it is byte-inert when disabled.

The design. A tiny CPU MLP (`overlay.rs`: 16 inputs → 12 hidden, tanh → 6 outputs; hand-rolled, no NN crate, no dependencies, pure deterministic f32) reads the situation the appraisal already computes (drive levels, affect, health, threat, enclosure, mortality, grief, winter, carrying) and emits **bounded biases on the drive arbitration**, scaled by an `nn_modulation` gene — disabled ⇒ bias is exactly 0.0 ⇒ instinct byte-identical. It **learns in-life** by reward-modulated three-factor Hebbian plasticity, `Δw = η·r·pre·post`, where reward `r` is the change in the mind's *own* well-being (drive satisfaction + health + valence, dimmed by grief) — an intrinsic, deterministic signal with no external supervision; weights and reward are clipped. The genome carries only the *learning machinery* — `nn_enabled` (g25), `nn_learn_rate` (g26), `nn_modulation` (g27) — not the weights themselves: an **indirect encoding** in which the germline evolves *how to learn* while lifetime Hebbian plasticity does the adapting. This is the **Baldwin effect** (Baldwin 1896; Hinton & Nowlan 1987) realised in an evolved plastic network (Soltoggio et al. 2018), with the reservoir-computing intuition (Jaeger 2001; Maass et al. 2002) that a fixed random substrate plus a light learned read-out can be expressive. The overlay nudges the same drive arbitration that the appraisal/affect machinery (OCC, Ortony, Clore & Collins 1988; EMA, Marsella & Gratch 2009; FAtiMA, Dias et al. 2014) already drives.

The results, told straight.

- *It genuinely learns in-life.* AC47 (harsh world, 600 ticks, `nn_enabled` ablation): with the overlay on, 6/6 agents form overlays and their total weight magnitude moves over a life ($\Sigma|w|$ 318.05 → 518.46) and stays finite; the gene-off control is inert ($\Sigma|w| \approx 0$, instinct byte-identical). The mechanism is real and harness-safe. AC47 asserts learning, **not** a fitness win.
- *It slightly hurts where instinct is tuned.* The A/B (`overlay_ab`, identical showcase genome, instinct vs overlay, 24 seeds × 800 ticks, harsh world) gives scalar 0.682 (instinct) vs 0.672 (overlay), **Δscalar −0.010**, Δsurvival −0.036 — the overlay **hurts**. The showcase instinct is already well-tuned for harsh-world survival, so a randomly-initialised overlay mostly injects early-life noise that one ~800-tick life of Hebbian learning cannot fully recover.
- *Evolution itself rejects it.* Letting evolution choose (`overlay_evolve`, 40 independent searches over the full 28-gene genome, harsh world; the overlay can be ablated for free), `nn_enabled` is selected in ~18% of champions versus ~5% for the *known-rejected* quantum core — a fair, apples-to-apples comparison since both start OFF in the carried baseline and are mutated by the same per-gene rule. Against a 50% random null this is a clear rejection (one-sided binomial $p \approx 2\times10^{-5}$, $n = 40$). Among the few champions that keep the overlay, evolution shrinks its influence to modulation ≈ 0.22 — even when retained, dialled down to a faint nudge. *(The `empowerment`/`imagination` selection rates, ~55%/70%, are **incumbent-ON** references reflecting that prior, **not** a clean 50% null — the load-bearing comparison is nn vs quantum vs the 50% line.)*

The honest conclusion. A learned, lifetime-plastic, gene-gated neural overlay is **feasible and harness-safe** — it integrates without breaking a proof and it demonstrably learns — but it **does not earn its keep where instinct is already well-tuned**: it loses the A/B and selection turns it off. The defensible reading is that learning has headroom only where instinct is *not* pre-optimised; testing the overlay on a novel or shifting task the hand-built faculties were never tuned for is the open next experiment, not a result we claim here. We report the null because it is a finding.

4. Methodology: believability as a falsifiable measurement

We contend the right way to make "the NPC feels alive" scientific is to **pre- register machine-checkable proxies** and let them gate development. Each criterion is one of:

- a **controlled experiment** (e.g. teach an affordance, then test behaviour),
- an **ablation** (two architecturally identical, same-seed agents differing in one mechanism or in lived experience — the difference *is* the effect), or
- a **multi-seed central tendency** (median over seeds) for emergent, seed-sensitive multi-agent phenomena.

The harness (`cargo run -p daimon-game --example believability`) runs all criteria and exits non-zero if any regresses. This caught real bugs the authors would not have seen by eye (e.g. an invented heal-goal hijacking decisions during starvation; stale-engagement mis-attribution in affordance learning). The harness is, we argue, the reusable contribution: a template for empirical game-AI claims.

4.5 Formal properties (machine-checked theorems)

Believability is measured; the architecture's *mechanisms* are also **proved**. We state nine theorems about the implemented system and pair each with a check that verifies the claim **on the code** (`cargo run -p daimon-game --example proofs`); full proofs are in `PROOFS.md`. The pairing is the methodological point: a theorem counts only when its written proof is matched by a green machine-check, and the checker exits non-zero on regression (a code change that breaks a proved property turns the proof red). During authoring this caught an overclaim — "TFT wins the field" was actually a *tie* with Grim (T8), corrected to *tied-optimal*. Two unit tests guard the mechanisms added in §3.21–3.22: `walls_block_predator` (a placed wall actually occludes the stalker, so shelter is protective not cosmetic) and `death_removes_mind_from_living` (permadeath truly removes the agent).

#	Property	Theorem (informal)
T1	**Determinism**	`(seed, genome)` ⇒ a unique trajectory; the AI is a pure function of its seed (SplitMix64 + fixed program order).
T2	**Homeostatic boundedness**	every drive stays in `[0,1]` and every learned bias in `[0.35,2.5]` — an invariant under all mutators.
T3	**Homeostatic stability**	curiosity is a geometric contraction to setpoint `0.25` with Lyapunov rate $(1-\alpha)^2=0.9025$; globally asymptotically stable.
T4	**Evolutionary elitism**	best-so-far fitness is monotone non-decreasing across generations.
T5	**Convergence**	elitism (T4) ∧ mutation $\sigma \geq 0.02 > 0$ ⇒ a.s. convergence to the optimum (Rudolph 1994), hypotheses verified on the engine.
T6	**Bell-CHSH / Tsirelson**	the quantum-cognition layer respects $\lvert S\rvert \leq 2\sqrt{2}$ for all states, $\lvert S\rvert \leq 2$ for separable states, and attains $2\sqrt{2}$ on the Bell state.
T7	**Self-organised criticality**	$\sigma = 1$ is an attracting fixed point of the SOC tuning map; the net self-tunes to criticality from both regimes.
T8	**Reciprocity non-exploitation**	tit-for-tat is exploited by at most $T-S$ and is tied-optimal (no strategy strictly outscores it).
T9	**Autonomous evolution**	the loop improves the *real* AI over baseline with no human input, halting on a principled verdict (the full-autonomy / evolves-over-time leg).

Two theorems are honestly scoped: T5's conclusion is Rudolph's theorem applied to machine-verified hypotheses (not reproven from scratch), and T9 is an empirical property (the loop *does* improve the real AI) rather than a closed-form result. Together with the novel architecture (§2–3), the autonomous self-improvement loop (§5), and these proofs, the system is a *novel, mathematically-proved, fully autonomous game AI that evolves over time*.

5. Results

All criteria pass deterministically (representative measured values):

#	Claim	Measure
AC1	Situational, non-repetitive thought	86% unique · top 2% · 100% grounded
AC2	Surprise = learned prediction error	mean 0.06, std 0.12; first-sight ≥3×
AC3	Derived insights + danger avoidance	10+ insights; taught 0 vs untaught 152 visits
AC4	Dialogue transfers actionable info	told→navigates, distance 24→1
AC5	Theory of mind beats chance	48% vs ~17%
AC6	Long-horizon projects	completed; persistence 56 ticks
AC7	Emergent information spread	median 3/6 agents reached

#	Claim	Measure
AC10	Hebbian association + cued recall	assoc 6.0 vs 0; recall ranks correctly
AC11	ACT-R base-level retrieval	frequent+recent ▯ one-off; decays
AC12	Risk-balanced, resilient decisions	safe-food choice; typical streak 133
AC13	Non-repetitive multi-act dialogue	124 distinct · 4 acts · top 4%
AC14	Concept genesis	5 things → 4 concepts; novel → +1
AC15	**Acting on the unforeseen**	**learned agent reaches healer (26→1, invented×40); naive does not**
AC16	Learned forward model	12 walls learned, 100% prediction
AC17	Empowerment shapes behaviour	escapes dead-end 16 vs 26 ticks (ablation)
AC18	Replay consolidation	replay-on activation > replay-off
AC19	Persistent minds	~3 KB round-trip, identical decisions
AC20	Imagination (planning)	reaches food behind a wall; reactive fails
AC21	Meta-motivation	self-revises curiosity weight 1.0→0.35, re-ranks arbitration
AC22	Quantum order effects	TV(A·B, B·A) = 0.205 (>0.05); classical = 0
AC23	Quantum interference	P_quantum 0.00 vs P_classical 0.50; interference −0.50
AC24	Quantum-decision agent	goal distribution shifts with deliberation order: TV 0.202
AC25	Self-organised criticality	branching ratio self-tunes σ 0.40 → 1.00 (edge of chaos)
AC26	Dynamic range peaks at $\sigma \approx 1$	Δ: sub 17.3 < crit 24.4 > super 18.0 dB
AC27	Self-improvement	evolved beats hand-tuned baseline (+0.086 scalar), best-so-far monotone
AC28	Self-evaluation + honest halt	self-halts with a Verdict; learns gene sensitivities
AC29	Anticipatory homeostasis	foresight ablation: critical-need time 31.9% → 23.5%
AC30	Commons-aware foraging	yield/disperse ablation (fair world): 11.3% → 5.1%
AC31	Conceptual entanglement	CHSH S = 2.828 (>2, = Tsirelson $2\sqrt{2}$); separable control 0
AC32	Entanglement entropy	Bell = ln 2; separable = 0; rises monotonically with binding
AC33	Learning progress	forward-model error 1.00 → 0.33 as world is learned; peak LP 0.92
AC34	Cumulative culture	affordance spreads peer→peer w/o contact; false memes corrected by experience (LP gate)
AC35	Learning-progress curiosity	engages the learnable; not fooled by unlearnable noise (LP 0.00 vs novelty 0.85)
AC36	Stigmergy (ACO)	colony self-organises onto the short route 100% vs 50% no-trail control
AC37	Stigmergy in the live world	worn paths emerge on foraging corridors (top-5% holds 100%); zero without
AC38	Scale generalisation	trained policy holds across village size 6→9% · 12→4% · 18→3% critical-need
AC39	Affect (circumplex)	safe+fed reads "content"; predator+harm reads "afraid" (valence↓, arousal↑)
AC40	Affect modulates behaviour	contentment loosens curiosity 0.25→0.62; fear→caution wired
AC41	Reciprocity (iterated PD)	tit-for-tat tops the tournament (499) > naive cooperation (450)
AC42	Emergent shelter (building)	`can_build` ablation: ON shelter-goals 49, builds 49, walls 49 (all >0); OFF control 0/0/0

#	Claim	Measure
AC43	Mortality / fear of death	mortal vs immortal twin: dread 1.00 (off 0.00); valence +0.15<+0.59; arousal 0.66>0.26; TMT-defensive 51>28 ticks
AC44	Grief over a bonded peer	bonded loss: grief 0.78, mourn 84 ticks; stranger: grief 0.00, mourn 0 (asymmetry)
AC45	Grief resolves (dual-process)	oscillates (mourn 187 / restore 204); decays to 0.22 alone; support speeds it (mourn 67<187)
AC46	Winter provisioning	`can_provision` ablation (3 seeds, full winter): ON cache peak 69, 7 survive; OFF cache 0, 2 survive
AC47	Neural overlay learns in-life	`nn_enabled` ablation: ON 6/6 overlays, Σ\
●	**End goal reached**	loop returns `ReachedTarget` in 3/5 searches; champion clears every facet at once; held-out 2/5 on unseen seeds (survival 0.88) — real but seed-sensitive

Plus AC8 (LLM-deliberator seam: offline contract test + `--features llm-http`), 82 unit tests, clippy-clean, native + WebAssembly builds, and the nine machine-checked theorems of §4.5 (`cargo run -p daimon-game --example proofs`).

5.2 System 2 — the learned overlay, evaluated honestly

The neural overlay (§3.24) is the architecture's first neural net, and we report its evaluation as a **null result** rather than a win. Three experiments, all deterministic and reproducible:

A/B — instinct vs overlay (`overlay_ab`, identical showcase genome, 24 seeds × 800 ticks, harsh world):

Policy	scalar	survival
instinct	0.682	0.377
overlay	0.672	0.341
Δ	**−0.010**	**−0.036**

The overlay slightly **hurts** in a domain instinct already masters — a genuine negative result.

Evolution chooses (`overlay_evolve`, 40 independent searches over the full 28-gene genome, harsh world; the overlay is freely ablatable):

Faculty	Selected in champions	Prior
`nn_enabled` (the overlay)	**18%**	OFF (incumbent)
`quantum` (known-rejected)	5%	OFF (incumbent)
`empowerment` (upper-reference)	55%	ON (incumbent)
`imagination` (upper-reference)	70%	ON (incumbent)

Evolution leans clearly against the overlay (18% vs a 50% random null; one-sided binomial $p \approx 2\times10^{-5}$, $n = 40$). The apples-to-apples comparison is `nn_enabled` (18%) against the known-rejected `quantum` (5%) — both start OFF and are mutated by the same rule; the ON-incumbent `empowerment`/`imagination` rates are soft upper-references, not the null. Among the few champions that keep the overlay, modulation shrinks to ≈ 0.22 (mean champion scalar 0.679, mean 10.2 generations). The honest arbiter agrees with the A/B: where instinct is well-tuned, a learned overlay does not earn its keep.

5.1 Benchmarks — evolvability, performance, generalisation

A dedicated suite (`cargo run -p daimon-game --example benchmark --release`) reports the headline numbers below. All cognition is local deterministic Rust — no GPU, no network, no ML libraries, no external model weights (the optional learned overlay of §3.24 is a tiny hand-rolled CPU MLP, default off in these benchmarks) — so the only machine-dependent figures are the wall-clock throughputs (measured on an Apple-silicon laptop).

Performance (raw throughput of the full cognitive cycle):

Setting	Throughput
1 agent	~212,000 cognitive ticks/s
6-agent village	~37,700 ticks/s (~226,000 agent-ticks/s)
18-agent crowd	~6,300 ticks/s (~114,000 agent-ticks/s)
Fitness evaluation	~13.0 ms per genome (a full 600-tick, 6-agent life) → ~77 genomes/s
A whole serialised mind	~2,095 bytes of JSON

At ~226k agent-ticks/s a single core runs a six-agent village far faster than real time; the self-improvement loop evaluates ~77 whole lives per second.

Evolvability (5 independent searches from different seeds, full fitness budget): baseline (hand-tuned) scalar 0.757; every search evolves a champion that beats the hand-tuned baseline (mean scalar gain **+0.061**, mean ~4.7 generations of search). A majority — **3/5 searches** — reach the full 7-facet end goal, and a fraction — **2/5 champions** — still meet that strict bar on a held-out set of unseen seeds. Generalisation is real but seed-sensitive: the loop reliably *improves on* the baseline, while clearing all seven facets at once remains the harder, noisier target. (Strict end-goal success is lower than the v1.0 report because the genome grew to `N_GENES = 28`; the extra genes are inert in the fair world and dilute the search, so a given budget reaches the all-facets bar less often.)

Zero-shot generalisation (tasks and worlds never trained for):

- **Acting on the unforeseen** (Praxis goal-genesis): an agent that merely *lived beside* a secretly-healing form crosses the map to it when hurt — pursuing a goal in no drive, planner, or goal table, for an entity type the architecture was never coded around. Across 8 seeds, the experienced agent reaches the healer **8/8** while an identical inexperienced control ignores it **8/8** — the *only* difference is lived experience.
- **Unseen village sizes** (champion tuned on 6 agents): critical-need time stays low across a 3× range of population — 6 (2.9%), 10 (5.3%), 14 (9.4%), and 18 (2.6%) agents all hold — so the evolved policy generalises across crowd density.
- **Unseen world layouts**: the champion's aggregate believability generalises (scalar ~0.79 averaged over five unseen maps); the strict *all-seven-facets-at-once* bar is harder and shows honest run-to-run variance — averaged over the five maps the full 7-facet target is *not* met, and per single world only 2/5 clear all seven facets at once (one facet typically dips). Generalisation in aggregate scalar is solid; the strict all-facets bar is noisier and not reliably cleared per world.

6. Discussion

What is genuinely new for game AI. The combination of (a) self-invented ontology *grounded in learned affordances* rather than designer labels, (b) empowerment as a live drive, (c) imagination over a learned model, and (d) self-revised values — under one deterministic, persistable, ablation-tested roof — is, to our knowledge, not present in shipping or published game-AI systems. AC15 in particular is a demonstration of open-ended autonomy that cannot be reproduced by templates or hand-rules: the only difference between the agent that exploits the novel healer and the one that ignores it is *experience*.

Honest limitations. This is not general intelligence. Perception is a structured percept, not pixels; the world is a grid; the System-2 reasoner is an offline heuristic in the tested build (the LLM seam is wired and contract-tested but not run live here); empowerment and the forward model are exact/tabular, not learned representations; meta-motivation revises drive *weights*, not the drive *set*. The quantum-cognition layer is quantum *probability* simulated on a CPU — a descriptive model of human non-classical judgment, **not** a quantum brain or a consciousness claim (Busemeyer & Wang 2015). The criticality substrate (§3.13) is verified as a standalone mechanism — the self-organising controller and the dynamic-range result are real and reproduced — but it is not yet wired into the live agent's salience gain; that integration is next, not done. The harness measures specified proxies, not believability in full generality. These are the next rungs, not refutations.

A genuine negative: learning does not beat tuned instinct (here). The most important new result is one we report against our own interest. The learned neural overlay (§3.24, §5.2) — the architecture's first neural net — *does* learn in-life and integrates without breaking a proof, but in the harsh world where instinct is already well-tuned it slightly **hurts** (Δscalar −0.010) and evolution selects it off (18% vs a 50% null, p ≈ 2×10^{-5}). We take this as a real finding, not a setback: a learned mechanism is feasible and harness-safe, but it earns its keep only where instinct is *not* pre-optimised. Whether a regime instinct cannot pre-solve flips that verdict is the open next experiment. We deliberately do **not** claim "learning improves the minds."

Future work — now chosen by the machine, not the authors. The autogenesis loop (§3.14) has localised the open frontier: *survival* is the one facet no parameter setting reaches, so the next mechanism is a **foraging / active- inference planner** (expected-free-energy action selection) aimed squarely at it, after which the same self-improving loop re-searches the enlarged space. Beyond that: live LLM deliberation; planning *with imagination as rollouts*; meta- motivation over the drive *set* (inventing new drives, not only reweighting); wiring the criticality substrate into the live salience gain; learned (non- tabular) world models and empowerment; pixel perception (SIMA-style); and multi- agent cultural evolution (Project Sid scale). The loop is the permanent engine; these are its fuel.

6.1 Threats to validity

We name the threats so a reader can weigh them.

- **Construct validity (does the harness measure believability?).** The 47 criteria are *proxies* — survival, decision balance, dialogue variety, emotional responsiveness, mortality salience, grief, winter provisioning, etc. — not human ratings. We make no claim that passing them equals being judged "alive" by a player; we claim only that each proxy is a necessary, ablation-isolated, falsifiable signal, and that the set is broader than any prior game-AI evaluation we know. A human study is future work.
- **Scope of the learned-overlay result (single domain, short lives).** The neural overlay's null (§5.2) is measured in *one* harsh domain over ~800-tick lives where instinct is pre-optimised. We do **not** generalise it to "learning cannot help"; the result is scoped to a mastered domain at this horizon, and a novel/shifting task or longer lives could plausibly flip it. We report the negative within those bounds and name the boundary as the open experiment.
- **Internal validity.** Determinism (Theorem T1) removes run-to-run confounds: every ablation compares architecturally identical, same-seed agents differing in exactly one mechanism, so a measured difference *is* that mechanism's effect. The risk is seed-specific artefacts; we mitigate with multi-seed medians for seed-sensitive emergent phenomena and with held-out validation for the evolved champion.

- **External validity (generalisation).** Perception is a structured percept on a 40×26 grid, not pixels; results may not transfer to high-dimensional perception. We report honest per-world variance (§5.1): aggregate believability generalises to unseen layouts and a 3× population range, but the strict all-facets-at-once bar shows run-to-run noise on individual unseen worlds.
- **Scope of the proofs.** T1–T8 are proved and machine-checked on the implementation; T5's conclusion invokes Rudolph (1994) over verified hypotheses rather than reproving convergence, and T9 is an empirical property (the loop *does* improve the real AI), not a closed-form theorem. The quantum-cognition and criticality results are *descriptive* models computed on a CPU, not physical claims (Busemeyer & Wang 2015).
- **Throughput figures** (§5.1) are single-machine wall-clock numbers (Apple-silicon laptop) and are illustrative, not benchmarked across hardware.

7. Related work

Generative Agents (Park et al. 2023); Voyager skill libraries (Wang et al. 2023); ReAct / Reflexion / Tree-of-Thoughts (Yao 2023; Shinn 2023; Yao 2023); BDI (Bratman 1987; Rao & Georgeff 1995); dual-process AI (Kahneman 2011; Booch et al. 2021); intrinsic motivation (Schmidhuber 2010; Pathak et al. 2017); **empowerment (Klyubin, Polani & Nehaniv 2005; Salge, Glackin & Polani 2014)**; classic unified cognitive architectures (Soar, Laird, Newell & Rosenbloom 1987) and **ACT-R activation (Anderson et al. 2004)**; free-energy / active inference (Friston 2010); model-based planning (MuZero, Schrittwieser et al. 2020; MCTS survey, Browne et al. 2012); machine theory of mind (Rabinowitz et al. 2018); GOAP (Orkin 2006) and HTN (Erol et al. 1994). **Quantum cognition** (Busemeyer & Bruza 2012; Pothos & Busemeyer 2009; order/QQ-equality, Wang, Solloway, Shiffrin & Busemeyer, PNAS 2014; concept combination, Aerts & Sozzo) — with the explicit descriptive-not-physical caveat (Busemeyer & Wang 2015). **Cross-disciplinary foundations**: relational quantum mechanics (Rovelli 1996), "it from bit" (Wheeler 1990) and constructor theory (Deutsch & Marletto 2015) for the information-first, relational stance; neural criticality / edge-of-chaos (Beggs & Plenz 2003; Chialvo 2010) for the operating-regime argument. **Self-improvement**: evolution strategies and the 1/5th-success rule (Rechenberg 1973); novelty and open-endedness (Lehman & Stanley 2011; POET, Wang et al. 2019); quality-diversity (MAP-Elites, Mouret & Clune 2015); and AI-generating algorithms (Clune 2019), of which autogenesis (§3.14) is a small, falsifiable instance — search over a cognitive genome with the believability harness as the objective. **Foraging & homeostatic decision** (§3.15–3.16): homeostatic reinforcement learning (Keramati & Gutkin 2014, *eLife* 10.7554/eLife.04811); the marginal value theorem (Charnov 1976, 10.1016/0040-5809(76)90040-X) and learned opportunity-cost-of-time (Constantino & Daw 2015, 10.3758/s13415-015-0350-y); survival-weighted foraging value (Mangel & Clark 1986, 10.2307/1938669; McNamara & Houston 1986); and congestion-game dispersion for the commons (Rosenthal 1973, 10.1007/BF01737559). **Developmental, social & affective** (§3.18–3.19): learning progress / intelligent adaptive curiosity (Oudeyer, Kaplan & Hafner 2007, 10.1109/TEVC.2006.890271; Baranes & Oudeyer 2013, arXiv:1301.4862); cumulative cultural accumulation in learning agents (Boyd & Richerson 1985; Cook, Lu, Hughes, Leibo & Foerster 2024, arXiv:2406.00392); stigmergy / ant colony optimization (Grassé 1959; Dorigo & Stützle 2004); the circumplex model of affect (Russell 1980, 10.1037/h0077714) and appraisal / action readiness (Scherer; Frijda 1986). **Reciprocity** (§3.20): the iterated prisoner's dilemma and tit-for-tat (Axelrod & Hamilton 1981; Axelrod 1984), reciprocal altruism (Trivers 1971), and indirect reciprocity (Nowak & Sigmund 1998). **Formal foundations** (§4.5): elitist evolutionary-algorithm convergence (Rudolph 1994); the CHSH inequality (Clauser, Horne, Shimony & Holt 1969), its quantum (Tsirelson) bound (Cirel'son 1980), and Bell's theorem (Bell 1964); self-organised criticality (Bak, Tang & Wiesenfeld 1987); and the splittable PRNG

(Steele, Lea & Flood 2014). **Mortality, fear of death & grief** (§3.22): attachment theory (Bowlby 1969); the dual-process model of coping with bereavement (Stroebe & Schut 1999); terror management theory (Greenberg, Pyszczynski & Solomon 1986; meta-analysis Burke, Martens & Faucher 2010); a recent computational model of grief (Dulberg, Dubey & Cohen 2026), which we cite to *avoid* claiming the first such model; and instrumental self-preservation as a contrast to affective fear-of-death (Omohundro 2008). **Open-ended worlds for AI** (§3.23): the games-for-AI-research survey (Hu, Zhao, Wang, Du & Liu 2024) and MineRL (Guss et al. 2019). **Learned plasticity & emotion architectures** (§3.24): the Baldwin effect (Baldwin 1896) and how learning can guide evolution (Hinton & Nowlan 1987); evolved plastic neural networks (Soltoggio, Stanley & Risi 2018); reservoir computing (Jaeger 2001; Maass, Natschläger & Markram 2002); and appraisal-based emotion architectures (OCC, Ortony, Clore & Collins 1988; EMA, Marsella & Gratch 2009; FAtiMA, Dias, Mascarenhas & Paiva 2014). Full bibliographic detail in §10 (References) below.

8. Reproducibility

Deterministic by construction (one seeded SplitMix64 PRNG; no wall-clock, no threads in cognition). Reproduce the entire evaluation:

```
cargo run -p daimon-game --example believability --release   # all 47 criteria
cargo run -p daimon-game --example proofs        --release   # the 9 machine-checked theorems (§4.5; PROOFS.md)
cargo run -p daimon-game --example autogenesis   --release   # the self-improvement loop
cargo run -p daimon-game --example benchmark     --release   # evolvability/perf/zero-shot (§5.1)
cargo run -p daimon-game --example overlay_ab    --release   # System-2 A/B: instinct vs learned overlay (§5.2)
cargo run -p daimon-game --example overlay_evolve --release  # evolution chooses for/against the overlay (§5.2)
cargo run -p daimon-game --example study         --release   # render-free behavioural field study
cargo test                                                   # 82 unit tests
cargo run -p daimon-game --release                           # watch the village (3-D isometric)
```

Every numeric claim in this report is emitted by one of these commands; the `proofs` and `believability` harnesses exit non-zero on any regression, so the report's claims are continuously re-verifiable rather than asserted once. The architecture, mechanisms, and evaluation harness are original work (clean-room provenance in `PROVENANCE.md`); the intellectual lineage is the public literature cited in §10.

9. Conclusion

We set out to make a game agent that **authors its own ontology, goals, world model, and values** — and to make that claim *decidable* rather than rhetorical. Daimon composes its mechanisms on a deterministic, dual-process BDI spine — and now also gives the agent a felt need for shelter, an awareness of its own mortality, grief over a bonded peer, an open-ended seasonal world to provision against, and a learned neural overlay we evaluate honestly (and find does not beat tuned instinct); a forty-seven-criterion ablation harness turns "the NPC feels alive" into a battery of falsifiable, reproducible tests; an autogenesis loop makes that harness its own fitness function and improves the architecture with no human in the inner loop, reaching a pre-registered end-goal target that is *earned* (a reactive policy fails) and *held-out-validated*; and nine theorems — determinism, homeostatic boundedness and Lyapunov stability, evolutionary elitism and convergence, the Bell–CHSH/Tsirelson bounds, self-organised criticality, and reciprocity non-exploitation — are **proved and machine-checked against the implementation**. The result is, to our knowledge, the first game-AI architecture that is at once novel, autonomous, self-evolving, *and* mathematically proved on the code that runs. It is not general intelligence (§6, §6.1); it is a small, honest, and reproducible step toward agents that are minds rather than puppets — and, just as importantly, a template for proving and measuring such claims rather than asserting them.

10. References

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